

Native bridges across exact mathematics

Rational roots, nonenumerating populations and marked Picard-Lefschetz actions

Original-source rational root completeness, huge affine residue counts, general bilinear transvection laws and checked signed-cycle-word execution through genus four.

12

reusable native theorem
declarations

48

generated declarations audited

21

source-bound producer packets

Two reusable Lean modules and four public native certificate operations. The complete repository archive preserves the concurrent flavor advances and the prior research receipts.

Source, corpus receipts and reproduction commands accompany the repository snapshot.

The new connected layer

This 6 October 2026 continuation connects three existing producers to native mathematical definitions. Rational-root packets now emit complete original-source root theorems, finite affine residue domains emit source-equivalence and nonenumerating count theorems, and marked branch braids emit native signed-word execution and integral symplectic identities. Twelve reusable theorem declarations and forty-eight generated declarations are audited in this edition. The complete archive retains all tracked content and the parallel flavor session's changes.

The reproduction targets are `make native-bridges-receipts`, `make native-bridges-lean` and `make test`. The receipt target regenerates twenty-one source packets, their four combined Lean audit files, and four cold-service examples. The Lean target builds the two new modules, prints the axioms of the twelve reusable theorems, runs namespace lint and compiles all twenty-one packets. Emission is explicitly execution-unverified until the source is compiled. Standard axioms are permitted; `sorryAx`, native evaluation axioms and extra analytic premises are not used by the audited declarations. This focused build does not refresh the historical whole-repository census.

Rational roots retain the original polynomial

The existing rational-root producer normalizes a nonzero polynomial f over $\mathbb{Q}$ and clears denominators. If n is its degree and D clears every normalized coefficient, the scaled integral polynomial has coefficients $g_i = (f_i/f_n)D^{(n-i)}$. Its leading coefficient is one. The mathematical native completeness theorem already connected g 's integer roots to f 's rational roots by z/D , but the existing packets stopped at Python replay.

The new emitter constructs f from the original coefficients, including their original leading coefficient. It proves its degree, nonzero leading coefficient, monicity of g and the entire polynomial scaling identity in Lean. Coefficient extensionality handles indices at most n individually; the proved degree bounds handle every larger index. It then checks the concrete output against `NativeRationalRoots.roots` by kernel reduction and applies `complete_nonmonic`. The final theorem quantifies over every rational r :

`f.eval r = 0` iff r belongs to the checked finite rational root set.

The existing signed-remainder Sturm packet is replayed at the producer boundary, but its real-root variation argument is not trusted by the native theorem. Lean independently evaluates the proved signed-divisor integer generator. The source identity and complete output are therefore native even though a generic efficient Sturm formalization remains open. A degree, rational-bit and signed-divisor work budget bounds this alternative proof route. Large scaled constants may be inexpensive for the existing Sturm producer and still exceed the native divisor certificate budget; rejection does not return a partial completeness theorem.

Eight retained fixtures cover nonmonic rational coefficients, zero constant terms, repeated roots, nonzero constants, no rational roots, negative leading coefficients, mixed denominators and degree eight. For $f=6x^2-3/2$, Lean proves that the complete rational

root set is $\{-1/2, 1/2\}$. For $x^4 - x^2$ it proves $\{-1, 0, 1\}$, and for $(x-1)^3$ it proves $\{1\}$. Each packet has a namespace bound to the original coefficient list, so multiple different source certificates can coexist in one audit file. The bridge can now supply native cubic and division-polynomial root semantics to elliptic consumers; it does not yet prove the Python point-law interpreter or every halving packet correct.

The elliptic consumer now goes further. It completes the square for the supplied Weierstrass model, emits a native discriminant-nonzero proof for that completed model, and uses the checked cubic-root theorem with Mathlib's actual affine point group. The resulting theorem quantifies over every actual rational point and characterizes two-torsion membership in the complete list, including infinity. Three retained curve packets cover $y^2 = x^3 - x$, $y^2 = x^3 - 2$ and the generalized coefficients $[1, 2, 3, -4, 5]$. The first has four rational two-torsion points; the other two have only infinity. Smoothness is proved in the generated source, not supplied as an unchecked packet flag. Generalized original-model coordinate transport and the Python point-law interpreter remain separate refinement tasks.

Large residue populations without interval enumeration

The new `ResiduePopulation.Cell` contains original-coordinate bounds, a positive modulus and a canonical residue. Its native values are exactly the integers x satisfying $lo \leq x \leq hi$ and $x \bmod m = r$. The lattice-coordinate theorem writes $x = r + mk$ and derives the exact quotient bounds, including negative endpoints. This gives an injective interval image and the count

```
max(0, floor((hi-r)/m) - floor((lo-1-r)/m)).
```

Finite unions of disjoint cells are counted by adding these short floor expressions. The native population is a deduplicated finite set, and its membership theorem remains an existential original-coordinate cell condition. The proof of its count never reduces that enormous finite set: it applies the lattice image theorem, injectivity and finite disjoint-union cardinality.

`residue_certificate` runs the existing exact semilinear compiler and emits a separate native source-equivalence theorem. In this edition the original atoms must be affine sign or remainder comparisons, combined through and/or/not. Lean proves the equivalence between that original source and the emitted cell union with `omega`, checks modulus/residue validity and every pair's disjointness, then verifies the count by short floor arithmetic. The Python compiler's assertion of completeness is not used as a premise. The restrictions keep the universal source proof within Presburger arithmetic; higher-degree sign cells still require native Sturm or another sign certificate.

The large retained source is $-10^{30} \leq x \leq 10^{30}$ together with $x \bmod 6 = 1$ and $x \bmod 9 = 4$. The compatible class is $x \bmod 18 = 13$, and its exact cardinality is 11111111111111111111111111111111. Lean checks source equivalence and that count without enumerating the interval. Other fixtures cover Boolean unions with exclusions, negative affine coefficients inside a remainder, an empty Boolean source and incompatible noncoprime residues. This closes a useful large-domain compiler bridge. General

polynomial cells, symbolic projected-value counts, optimizer transcripts and an implementation proof of arbitrary rank/select remain separate tasks.

Picard-Lefschetz actions become native algebra

For an alternating bilinear form B on a module V and a declared vanishing cycle d , define

$$T_d(x) = x + B(d, x) d.$$

The new Lean module works over any commutative ring. Alternation implies skew symmetry by expanding $B(x+y, x+y)$, without dividing by two. The module proves that T_d preserves B , that $x - B(d, x)d$ is its inverse when d is isotropic, that disjoint-cycle twists commute, and that cycles with $B(d, e)=1$ satisfy the braid relation $T_d T_e T_d = T_e T_d T_e$. The integer specialization is the algebra needed by marked surface homology. These proofs are independent of genus, numerical periods and a finite search for examples.

The matrix bridge defines the exact rank-one matrix $I + d(d^T J)$ and proves that its action equals the bilinear transvection for `Matrix.toBilin' J`. Thus the matrix formula used by the existing marked branch-braid engine is now connected to the general native operation. A native signed-word evaluator folds those matrices in the declared order, including inverse signs.

The braid emitter reconstructs every factor from the declared integral cycle and standard intersection matrix. It stores concrete intermediate matrices and checks one factor product at a time with `decide +kernel`, then rewrites those checked steps into the complete native word action. This avoids repeatedly expanding a long functional matrix product during proof reduction. The final integral symplectic equation $M^T J M = J$ is also kernel checked. Word length and total cubic matrix work are bounded. Twenty-one packets across the bridges include five marked-word examples through genus four, with an inverse word and an empty word.

The topological premise remains the declared marking and braid word. This edition does not infer branch braids from arbitrary polynomial parameter loops, identify an unmarked Smith basis with a symplectic basis, or identify a simplicial mesh with an algebraic quotient. Those geometric bridges remain open. The new native algebra can be reused when those identifications are supplied, and can support the existing reflection-kernel and higher-genus machinery without promoting an unrelated matrix realization to a geometric map.

Public operations and the remaining boundary

The JSONL service now exposes `native_rational_certificate`, `native_two_torsion_certificate`, `native_residue_certificate` and `native_braid_certificate`. Four cold catalogue requests reproduce the arithmetic root packet, a huge bounded residue population, actual elliptic two-torsion and a genus-three marked word. Registered elliptic objects also expose `native_two_torsion`. All results retain the emitted source, its hash and an execution-unverified marker until compilation. Parser/JSON-to-native refinement remains open; the compiled theorem refers to the generated mathematical source, which is explicitly printed in the packet.

This push narrows the original roadmap at three concrete joints. Efficient generic Sturm certificates are still needed to escape the signed-divisor budget and admit higher-degree sign cells. Elliptic point arithmetic and coordinate-integrality transport still need an interpreter refinement. Geometric branch-braid extraction, mesh marking and analytic period identification remain separate from the native transvection algebra. The following repository context records the broader arithmetic, curve, metric, Gamma, sequence and solver avenues that these new bridges can feed.

The connected repository context

Native roots become actual elliptic fibres

`NativeRationalRoots` turns the proved signed-divisor integer generator into a complete rational-root generator. A producer supplies an ascending integral coefficient list, a nonzero rational scale D , monicity and an exact coefficient identity with the scaled polynomial. Lean checks those premises. The integer-root output is obtained from `NativePolynomialRoots`; it is no longer supplied as a separately assumed complete list. Roots are transported back by z/D . A leading-coefficient normalization extends the result to every nonzero rational polynomial for which the checked scaled integral coefficients are supplied. Zero constant terms and repeated roots retain the existing recursive treatment.

`EllipticPointDivision` connects the completed-square model $y^2 = x^3 + Ax^2 + Bx + C$ to Mathlib's actual affine point group. For nonsingular affine points and nonzero 2 , two-torsion is equivalent to $y=0$; the x coordinate is therefore a cubic root. The x coordinate in Mathlib's doubling formula satisfies exactly the halving quartic proved in the previous push. A nonzero discriminant allows each rational cubic root to construct a nonsingular branch point. The torsion list includes infinity. Native rational roots then prove this list complete, and an actual checked anchor H with $2H=P$ gives all halves by translating that torsion list. The resulting theorem covers the entire fibre, including infinity and exceptional points through the group theorem; only the coordinate doubling equivalence uses its explicit nonzero-denominator premise.

The remaining elliptic obligations are now sharply located. The Python point-law implementation, JSON parser and certificate interpreter still need a formal refinement to these native definitions. Anchor discovery remains a producer task; its group equation is a

checked premise, not a rank assertion. The independence machinery still proves the stated lower bounds under its own certificate semantics; full Mordell-Weil rank determination, arbitrary n -division and generic higher-genus covering maps remain separate work. Generalized-model transport must continue to preserve original-coordinate integrality.

Compositional domains, CRT and nonmonotone images

`FiniteDomainCertificate` provides a transcript language for literal finite sets, unions, intersections, decidable filters and images. Its denotation retains the source predicate, and its evaluator produces a deduplicated finite set. A structural completeness theorem relates evaluation to denotation. Rank counts distinct accepted integers strictly below a value. Strict rank growth and selected-value uniqueness show that membership plus rank identifies one value. A source-equivalence theorem transports that guarantee back to a supplied original predicate. This is useful for nonlinear images because several source parameters can have one output value; counting source occurrences would give the wrong rank.

`GeneralCRT` removes a hidden coprimality restriction from the mathematical chart layer. A simultaneous pair of natural residue conditions exists exactly when the residues agree modulo the gcd. The canonical compatible anchor has an lcm-period chart and, for nonzero moduli, lies below that lcm. A separate integer theorem keeps the same chart exact for negative parameters and residues. The remaining compiler work is a universal parser/AST correspondence, generic Sturm-variation certificates, assembly of every emitted semilinear cell and a verified algorithm for rank/select and optimizer search. The new transcript theorem proves the meaning of assembled checked data; it does not claim that an arbitrary Python compilation already constructs those proofs.

The population implementation now supports tuple-key joins. Bounded value joins preserve every original pair and every multiplicity. Symbolic joins choose one complete curve relation and enforce the additional polynomial equalities through the curve predicate language. A test joins the square and linear fields over $2 \times 10^{30} + 1$ source parameters and retains exactly the diagonal without enumeration. Unsupported primary curve relations still fail explicitly. Arbitrary multivariate joins and a general complete relation solver remain open.

Nonmonotone projection now includes integer even quartics $ax^4 + bx^2 + c$. `QuarticCollision` proves that equal values have $x=y$, $x=-y$ or $a(x^2+y^2)+b=0$, and proves bounds for the circle branch. The Python backend handles the infinite reflection through source predicates and exhausts the finite circle when its work bound permits. It retains the least original accepted owner even on disconnected domains, then reuses the existing image/rank interfaces. Arbitrary nonsymmetric quartics and higher-degree nonmonotone collision curves still require new complete backends; a budget rejection returns no partial completeness claim.

Variable-width factorial arithmetic

FactorialWindow proves the p-adic valuation of $N!/(N-W)!$ as a difference of factorial valuations and then as two finite Legendre sums. At the native level, natural subtraction makes $W > N$ a truncated window; the public Python operation requires $0 \leq W \leq N$. A valuation not divisible by k proves that the positive window product is not a k -th power. The Python witness search uses the bit length of N and a bounded supplied prime list, so it can process widths that cannot be enumerated. Passing every tested prime produces UNKNOWN.

The retained generated theorem proves that $(10^{30})!/(10^{30}-10^{20})!$ is not a square. Its 2-adic valuation is 100000000000000000011, which is odd. Lean evaluates the finite floor sums and checks logarithmic truncation bounds through a power inequality; it does not reduce either enormous factorial. This closes a useful variable-width Gamma-ratio obstruction route. Complete classification of arbitrary variable-width products, sums or differences of factorials, general Gamma arguments and multivariate creative telescoping remain open. Sequence-definition review, continued-fraction execution at very large unit height, broad nonlinear recurrences, transform endpoints and Tauberian conclusions retain their previous boundaries.

Certified periods, inverses and marked monodromy

The analytic extension uses rational complex centres and an induced row norm with $|\operatorname{Re} z| + |\operatorname{Im} z|$ on entries. For a supplied enclosure $\|A - C\| \leq e$, exact Gaussian elimination computes C^{-1} . If $q = \|C^{-1}\|e < 1$, Neumann's lemma proves nonsingularity throughout the ball and bounds inverse error by $\|C^{-1}\|q/(1-q)$. Product balls use the full cross term. Integral recognition requires error strictly below one half, checks entry and row compatibility, and can check the standard symplectic form exactly. The general operation explicitly requires an independent integrality premise: closeness alone cannot establish integrality.

Long analytic paths now have an optional outward-rounded dyadic mode. Centres are rounded with a bounded row error added to the enclosure; the error bound itself is rounded upward. Certified disks also keep the exponential bound controlled. This prevents repeated rational matrix products from accumulating huge denominators while preserving the existing exact mode. The marked Legendre wrapper combines the original certified seed, closed continuation, inverse and product balls, and symplectic recognition. Its integrality justification is the supplied standard integral cycle marking, not numerical proximity.

The retained nontrivial loop starts at $1/2$, passes through $1/2+i/4$, $-1/4+i/4$, $-1/4-i/4$ and $1/2-i/4$, and returns to $1/2$. At Taylor order 24 and 128-bit outward rounding, 148 certified steps identify the cycle matrix $[[1,2],[0,1]]$. The row enclosure is approximately $2.22466826496 \times 10^{-9}$. This is a concrete certified marked monodromy computation. General singular endpoint asymptotics, arbitrary integral period markings, rigorous quadrature with arbitrary cycle inputs, holomorphic-basis normalization and the analytic estimates themselves in Lean remain open.

Curve structure, arithmetic reduction and global geometry

The existing exact curve engine retains its explicit quotient towers, two-isogenies, Richelot constructions, de Rham connections, superelliptic reductions, horizontal-section searches and selected Frobenius computations. The current push adds native elliptic fibres and a certified marked analytic continuation bridge. General higher-genus correspondences still need integral cycle maps, kernel degrees and singular-target treatment. Nonsplit stable reduction, arbitrary Puiseux root jets, arithmetic component twists, finite-field extension descent, cross-parameter p-adic Frobenius transport, full differential Galois groups, nonlinear idempotents and conjugate Hodge/Betti descent remain research fronts. A bounded ansatz search for horizontal sections cannot prove that every possible section has been found.

`cluster_graph_metric` turns the existing split-root geometric semistable graph into a global finite length metric and tropical cycle lattice. Exact Floyd-Warshall distances respect all graph identifications and parallel edges. A deterministic spanning tree gives integral fundamental cycles; every boundary is checked to vanish, the number of cycles matches the graph's cycle rank, and the edge-length Gram form has a positive determinant. A replay operation binds the entire packet to the original rational roots and odd prime and rejects changed source data or witnesses. These distances and cycle lengths are properties of the geometric reduction graph. They are not a complex-curve metric, a minimal arithmetic regular model or a Frobenius marking.

The remaining smooth geometry work includes a global quotient length metric on the analytic atlas, JSON-to-native geometric checker correspondence, global chart gluing and smoothness comparison, general period-normalized metrics and certified geodesics. The existing 108 closed metric boxes out of 144 leave 36 unresolved boxes; this push does not turn them into certified boxes. Numeric hyperelliptic periods remain scoped to their supported genera and numerical methods, and general cyclic-mesh cycle integration still needs a marked analytic bridge.

Solver transport and verification consumers

`iter_commands` accepts bounded text chunks and yields exact SMT-LIB command slices on demand. Its state handles comments, quoted symbols and doubled string quotes across every chunk boundary. It rejects malformed tails, top-level garbage and exceeded command or chunk limits. `solve_replay_stream` preserves native query barriers and batches only non-query transport. A query sink avoids retaining rows. Errors stop future queries while preserving completed earlier query results. Storage bounds cover the scanner and pending transport; they do not bound the solver's assertion state, the caller's input storage or an intentionally retained result list.

The scanner matched 277345 exact slices across 292 repository SMT files at a 127-character chunk size. A source-hashed replay of 16 repository and Why3 tasks reported no sat/unsat contradictions. Scanner allocation measurements used generated chunks and measured 1332 bytes for 1000 commands and 1340 bytes for 100000 commands. They are local measurements, not whole-process memory bounds. The earlier

external 181-file industrial cohort is not replicated by this receipt. Its reported throughput and solved-query gains retain their earlier provenance, and independent replication and broader throughput work remain necessary.

The Why3 bridge now allocates distinct names after sanitization, reserves generated local names and language keywords, safely encodes source-symbol comments, caps expanded powers and rejects unsupported free Boolean binders. These fixes eliminate concrete malformed or capturing translations. The original Why3 tool is unavailable in this environment, so its native rerun remains skipped. A universal SMT-to-Why3 semantic correspondence, SPARK/GNATprove VC correspondence, overflow and bitwise refinements remain open. Brainpool's residue-ring equation/export transport remains proved, while its 384-bit field primality and complete group implementation bridge still require a native prime certificate; trial factoring of $p-1$ leaves a composite 349-bit cofactor.

Global arithmetic and next research decisions

The historical descent coverage still records 79 classes: 15 local and 64 conditional, with zero unresolved coverage entries. Unit registration covers every currently needed obligation; this does not discharge the global Matveev-type premises. The 23 negative curves carrying those premises retain conditional theorem statements. The positive registry still records 73 closed cases and the earlier conditional entries. $k=22$ remains closed by its retained 100-chunk kernel check of 971659 source candidates. $k=94$ remains the next larger source with 5294219 candidates; no complete source theorem for it is claimed by this push. Generic native maximal-order computation, basis-independent nonmonic norm transport and recentered Skolem extraction remain separate from the rational-polynomial normalization delivered here.

Density-zero statements for general nonrigid branches, normalized Kummer/Riemann-Hurwitz statements and global Siegel/Boshernitzan ingredients remain mathematical obligations. Hall, Pillai and uniform integral-point conjectures remain outside the unconditional results. The bounded finite-domain emitter and general ordered weighted identity now have native closures. The next productive work is to refine elliptic execution and nonenumerating domain compilation, package the unordered-subset determinant form, extend certified marked periods to singular and higher-genus settings, and choose an arithmetic closure with an affordable source-proof budget. No roadmap item is promoted to a theorem merely because it now has a working data pipeline.